

**SOLANGE: Planning a Transdisciplinary Quantum-Oncology Research Platform Under
Technological Uncertainty**

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SOLANGE: Planning a Transdisciplinary Quantum-Oncology Research Platform Under Technological Uncertainty

This paper presents the project plan for SOLANGE, a research platform that combines quantum computing, high-performance computing, and artificial intelligence to study cancer mutations for which no approved targeted drug exists. The platform is the subject of a doctoral dissertation (Cohen, 2026a), and our team treats it here as a complex, transdisciplinary project.

Project Area: Quantum Computing for Non-Druggable Cancer Mutations

Lung cancer was the most frequently diagnosed cancer in 2022 and the leading cause of cancer death worldwide (Bray et al., 2024). The dissertation estimates that about 60% of non-small cell lung cancer (NSCLC) patients carry mutations for which no approved targeted therapy exists (Cohen, 2026a). Part of the obstacle is computational. Designing a drug for a mutated protein requires an accurate model of the electrons at the mutated site, and classical methods lose accuracy as the number of strongly interacting electrons grows. Quantum computers are expected to handle this problem better, but current hardware is noisy. IBM plans its first large-scale, fault-tolerant system, Starling, for 2029 (IBM, 2025a).

SOLANGE does not assume that quantum computation is needed. For each mutation, the platform runs two independent classical methods and decides whether the problem is classically tractable or whether quantum computation is warranted (Cohen, 2026a). A provenance layer called LEON seals every result and re-verifies it before anyone may rely on it, following the audit-trail logic of FDA 21 CFR Part 11 (Cohen, 2026a; U.S. Food and Drug Administration [FDA], 2003).

The project draws on chemistry, physics, quantum computing, mathematics, biology, oncology, software engineering, regulation, finance, and international health cooperation, and

each discipline defines success differently. It is also a wicked problem in the sense of Rittel and Webber (1972): the team cannot fully define what the platform must compute until it has tried to compute it.

Prior Work in This Space

What Went Wrong

Two patterns explain most failures in this space. The first is promises that ran ahead of evidence. MD Anderson Cancer Center spent \$62 million on an oncology project built on IBM Watson before canceling it, and a former IBM chief medical scientist said there was very little published evidence that Watson improved patient outcomes (Strickland, 2019). Watson also reasoned from statistics about main outcomes, while oncologists reason from individual cases, and the two sides never closed that gap (Strickland, 2019).

The second pattern is claims of quantum advantage that later moved. In 2023, IBM reported accurate results from a 127-qubit processor at a scale beyond brute-force classical computation (Kim et al., 2023). Researchers at the Flatiron Institute then reproduced the experiment with classical tensor-network methods, more accurately, and some of the simulations could run on a mobile phone (Schirber, 2024). For chemistry, Lee et al. (2023) found no evidence yet of an exponential quantum advantage across chemical space and advised that it should not be assumed. IBM itself wrote in late 2025 that quantum advantage had not yet been achieved, because rigorous validation was still missing (IBM, 2025b).

What Went Well and Why

Progress on hard cancer targets has come from patient, structure-based work. KRAS was long considered very difficult to drug because it binds GTP extremely tightly and had no known allosteric site. Ostrem et al. (2013) found a new pocket in the G12C mutant, and eight years later

the FDA granted accelerated approval to sotorasib for KRAS G12C-mutated NSCLC (FDA, 2021). In 2026, a Phase 1 trial of rezatapopt, which restores the normal shape of Y220C-mutated p53, reported a 20% response rate (The ASCO Post, 2026), and SOLANGE removed TP53 Y220C from its targets (Cohen, 2026a). Infrastructure advanced through long partnerships. Cleveland Clinic and IBM began a 10-year program in 2021 that combines quantum computing, artificial intelligence, and high-performance computing and trains the workforce the field lacks (Cleveland Clinic, 2023).

The lesson is direct. Projects in this space advanced when evidence set the pace, and they failed when marketing or a vendor roadmap set it. SOLANGE follows the first pattern. Every real biological target it has measured so far has proven classically tractable, and the platform reports that finding instead of hiding it (Cohen, 2026a). None of the projects above performs the task at the center of SOLANGE. The Cleveland Clinic and IBM partnership is the closest in spirit, but it builds shared research infrastructure. The dissertation found no counterpart, in the literature it reviewed as of September 2026, to a platform that decides per target, at runtime, whether quantum computation is warranted, by running two independent classical methods on the same active space and sealing each verdict in an audit trail aligned with FDA Part 11 (Cohen, 2026a).

How the Project Plan Was Constructed

The plan follows design science research, which builds and evaluates a working artifact (Cohen, 2026a). We built the plan from two directions. Working backward, the defense date fixed the end point. Working forward, we asked what each phase could deliver with the technology that exists today. The result allows about 20 months from proposal acceptance to defense, because simulator engineering, hardware access, and expert recruitment cannot be

shortened without losing rigor (Cohen, 2026a). Figure 1 shows the full timeline. The completed phases have already produced a working platform, a dissertation proposal, a research poster, and a manuscript prepared for arXiv submission (Cohen, 2026b). The Appendix shows screenshots of each.

Phases, Milestones, and Team Roles

Phases 1 and 2, the literature foundation and the deployed prototype, are complete. Phase 3A validates the architecture on a quantum simulator. Its 20-electron, 40-qubit run was completed on August 4, 2026, and a 24-electron run is pending. Between July and August 2026, small test jobs of 4 to 8 qubits also ran on three real IBM Heron devices (Kingston, Marrakesh, and Fez). These jobs validated the pipeline, not chemical accuracy. Phase 3B scales the run to IBM Heron r3 hardware from September to December 2026, if access is granted. Phase 4 evaluates the platform from January to April 2027 in six sub-phases, including a four- to six-member expert panel. Phases 5 to 7 cover writing, revisions, and the defense in late 2027 (Cohen, 2026a).

Four milestones anchor the plan: proposal acceptance, the Phase 3B access decision, the freeze of pre-registered evaluation criteria, and the defense. Three management tracks run beside the phases. The team re-baselines the roadmap each quarter against the largest calculation actually completed, a countermeasure from our A3 analysis. Manuel Blanco, a senior banker, leads funding and investor relations, including binational science grants and an academic collaboration with IBM Research Israel. Amit Mekel, a senior diplomat, leads international partnerships and health diplomacy with partner hospitals and health ministries. Doron Cohen leads the project and the technical work.

The Most Difficult Phase

Phase 3B is the hardest phase to execute, for three reasons. First, it depends on an outside decision, because access to Heron r3 runs through the IBM Quantum Network and can take three to six months to secure (Cohen, 2026a). Second, the hardware is immature, so a run can fail for reasons the team does not control. Our A3 found that earlier slippage came from gate dates that followed a vendor roadmap. Third, success is hardest to define here. A chemist may count a converged energy as success, a funder may expect a quantum result, and an oncologist wants clinical relevance.

The plan responds in three ways. Phase 3B is formally contingent: without access by December 2026, it becomes future work, and the defense proceeds on Phases 3A and 4 (Cohen, 2026a). A target enters the hardware stage only if it fits the largest calculation already completed on real hardware. Each discipline records its acceptance criterion before a run. Evidence discipline also matters. On September 23, 2026, the project retracted an energy result after a check showed that the mutation had not been applied to the structure (Cohen, 2026a).

Project Artifacts

Table 1 lists the artifacts the project would maintain if it went forward. The categories follow Section 4.6 of the PMBOK Guide (PMI, 2021).

Table 1
Artifacts the SOLANGE Project Would Maintain

Artifact	PMBOK category	How we would use it	Owner
Project charter	Strategy	Authorizes the project, names the four milestones, and states that Phase 3B is contingent on IBM access.	Doron
Roadmap	Strategy	Shows phases, milestones, and decision points (Figure 1). Re-baselined each quarter against measured capability, never against a vendor date.	Doron
Assumption log	Logs and registers	Records every modeling choice behind a result, such as the active space, selection threshold, and basis set, so each result can be traced to its assumptions.	Doron
Risk register	Logs and registers	Tracks IBM access, hardware noise, funding gaps, and expert recruitment, each with an owner and a planned response.	All
Stakeholder register	Logs and registers	Lists chemists, oncologists, the committee, regulators, funders, and foreign health partners, with what each group counts as success.	Amit
Change log	Logs and registers	Records formal changes, such as removing TP53 Y220C after rezatapopt entered trials and retracting an energy result.	Doron
Communications management plan	Plans	Defines who receives which result and through which channel. Every result carries its Cynefin context: established or experimental.	Amit
Test plan	Plans	Holds the pre-registered thresholds: 100% audit-trail integrity, at least 95% end-to-end execution, and an expert rating mean of at least 5.0 (Cohen, 2026a).	Doron
Budget	Baselines	Approved estimate for compute time, hardware access, and the expert panel. Compared with actual spending each quarter.	Manuel
Memoranda of understanding	Agreements	Set the terms of data use and expert participation with partner hospitals and research institutions.	Amit and Manuel

Project Management Model

We would manage SOLANGE with the Cynefin framework as the main model, inside a hybrid life cycle. Cynefin diagnoses cause-and-effect relationships in five contexts (PMI, 2021, Section 4.2.5.1). In a complicated context, the unknowns are known, and experts apply analysis

and good practices. In a complex context, the unknowns are unknown, so the team works in repeated cycles of probe, sense, and respond. When the context is unclear, the framework recommends breaking the problem into parts and matching each part to a context. SOLANGE needs this split. The classification stage is complicated, so it follows a predictive plan with fixed milestones. The quantum stage is complex, so it follows short cycles in which the team runs a calculation, measures the result, and decides what enters the next cycle. The quarterly re-baseline in Figure 1 schedules those cycles. Chapter 3 of the PMBOK Guide allows parts of one project to follow different approaches (PMI, 2021). This structure avoids both failures that Rittel and Webber (1972) describe: the fixed plan and free improvisation.

Cynefin leaves one gap, because Table 4-1 of the PMBOK Guide does not link it to stakeholders (PMI, 2021). We therefore add the cross-cultural communication model. It explains that the background of the sender shapes a message, and the background of the receiver shapes its interpretation (PMI, 2021, Section 4.2.2.1). A chemist, an oncologist, an investor, and a health official will read the same result differently, and the Watson experience shows the cost of such misreading (Strickland, 2019). Each result will therefore state its Cynefin context, established or experimental, and conditional results will be presented in a live briefing, not only in a written report (PMI, 2021, Section 4.2.2.2). Manuel applies this rule with funders, and Amit applies it with partner governments and hospitals. Every model costs time and expertise (PMI, 2021, Section 4.1), so the second model is justified only by the domain it adds. This combination is the tailoring decision at the center of the plan.

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Appendix: Supporting Artifacts

The screenshots below show the working SOLANGE platform and the dissertation behind the project plan.

Screenshot 1. SOLANGE platform, Dashboard: the classical tractability plane. Each point is a target classified by measured entanglement and active-space size (Cohen, 2026a).



Screenshot 2. SOLANGE platform, Orchestration: the two-method classifier (DMRG and SHCI) with sealed results for real targets (Cohen, 2026a).

LADDER - RUNG 3 — DMRG + SHCI last classical resort — the A/B/C classifier that decides whether quantum is even needed

THE CLASSIFIER — PDB TO A TWO-METHOD DECISION ⓘ

Part 1 — cluster (cached if this exact site was built before, otherwise built from scratch) · Part 2 — classify with both DMRG and SHCI, and report agreement

PDB ID (e.g. 8TDR) Chain (e.g. A) Residue # Expect (e.g. ARG) Target key (e.g. SDHB_C101Y)

AVAS criterion (optional — omit for per-radius auto-suggestion) 5.0,4.0,3.5,3.0 45 spin 0 **Classify**

CLASSIFIER RESULTS — METHOD 1: DMRG

DMRG A/B/C CLASSIFICATIONS · LAGUNA (SOLANGE_DMRG.PY) ⓘ **Refresh** **Export CSV** **Delete selected DMRG (0)**

TARGET	TIME	CAS	S_MAX	CLASS	RUNTIME	HARDWARE	(P8)LEON
TP53_C275F	2026-09-23 11:14	CAS(36,21)	0.79	B a (provisional)	1m42s	AMD EPYC 9554 ...	✓ 82ba299d6825f05e... Verify JSON Queue
TP53_R175H	2026-09-23 09:02	CAS(120,65)	1.31	B a (provisional)	527m52s	AMD EPYC 9554 ...	✓ 8b7430ea7b8f58d3... Verify JSON Queue
TP53_R175_NATIVE	2026-09-21 22:23	CAS(126,68)	1.08	B a (provisional)	228m55s	AMD EPYC 9554 ...	✓ 991491e6df4754ef... Verify JSON SHCI needs r
TP53_C275_NATIVE	2026-09-12 23:29	CAS(42,25)	0.78	B a	10m18s	AMD EPYC 9554 ...	✓ 45de5724bc35239b... Verify JSON SHCI needs r
TP53_C275F	2026-09-12 19:02	CAS(42,25)	0.80	B a	7m00s	AMD EPYC 9554 ...	✓ 55fd508fe9b551c0... Verify JSON SHCI needs r
TP53_C275_NATIVE	2026-09-12 13:06	CAS(42,25)	0.78	B a	9m49s	AMD EPYC 9554 ...	✓ 19773b359ee887b7... Verify JSON SHCI needs r
NEGCTRL_BORING	2026-09-09 13:14	CAS(66,55)	1.52	A a (provisional)	222m08s	AMD EPYC 9554 ...	✓ 87da163bd22c734d... Verify JSON SHCI needs r
FE2S2_PROXY	2026-09-02 14:25	CAS(46,32)	3.37	A a	54m45s	AMD EPYC 9554 ...	✓ fb6eb4aa834b55e2... Verify JSON SHCI needs r
TP53_C275F	2026-09-02 12:38	CAS(42,25)	0.68	B a	6m45s	AMD EPYC 9554 ...	✓ b3652ad84d87c887... Verify JSON SHCI needs r
TEST_DNMT3A_R882	2026-09-01 15:05	CAS(30,16)	0.20	B a	3.4s	AMD EPYC 9554 ...	✓ 2c42413330e16e89... Verify JSON SHCI needs r
DNMT3A_R882_AU...	2026-08-31 23:30	CAS(60,33)	0.66	B a	7m34s	AMD EPYC 9554 ...	✓ 27809c5d44312e67... Verify JSON SHCI needs r
DNMT3A_R882	2026-08-31 12:48	CAS(60,33)	0.66	B a	7m36s	AMD EPYC 9554 ...	✓ 17d7b0455a1e4fd3... Verify JSON SHCI n/a
TP53_R175H	2026-08-15 20:02	CAS(96,54)	0.78	B a	73m15s	AMD EPYC 9554 ...	✓ 259d146b4fe8e381... Verify JSON SHCI n/a

Screenshot 3. SOLANGE platform, Orchestration: pipeline test jobs on real IBM Heron hardware, July to August 2026 (Cohen, 2026a).

LADDER - RUNG 4 - QUANTUM bottom of the ladder — real IBM Heron QPU, where the classical rungs run out

QUANTUM HARDWARE - IBM HERON (REAL QPU) ¹ [Refresh](#) [Export CSV](#) [Delete selected QPU \(0\)](#)

TARGET	TIME	ACTIVE SPACE	QUBITS	EXACT HF (HA)	QPU (H) (HA)	ΔE	RUNTIME	
TP53_C275F (mutant) · toluene	2026-08-11 00:44	CAS(4,4)	8q	-269.642992 DMRG-SCF	-269.637350 QPU (H) (HF)	5.64 mHa · noise	2m05s QPU	i
TP53_C275F (mutant) · toluene	2026-08-11 00:23	CAS(2,2)	4q	-266.472783 DMRG-SCF	-266.464180 QPU (H) (HF)	8.60 mHa · noise	1m28s QPU	i
TP53_C275F (mutant) · toluene	2026-08-07 16:13	CAS(2,2)	4q	-266.472783 DMRG-SCF	-266.471125 QPU (H) (HF)	1.66 mHa · noise	21.0s QPU	i
STK11_LKB1 (native) · acetic_acid	2026-08-07 16:11	CAS(2,2)	4q	-224.797397 DMRG-SCF	-224.796959 QPU (H) (HF)	1.44 mHa · noise	21.0s QPU	i
KEAP1_LOF (mutant) · methanethiol	2026-08-04 12:15	CAS(2,2)	4q	-432.892979 DMRG-SCF	-432.891218 QPU (H) (HF)	1.76 mHa · noise	21.0s QPU	i
TP53_C275F (native) · methanethiol	2026-08-02 16:07	CAS(2,2)	4q	-432.892976 DMRG-SCF	-432.891207 QPU (H) (HF)	1.77 mHa · noise	21.0s QPU	i
CDKN2A_LOF (native) · guanidine	2026-08-02 14:50	CAS(2,2)	4q	-201.438640 DMRG-SCF	-201.436330 QPU (H) (HF)	2.31 mHa · noise	21.0s QPU	i
ARID2_LOF (native) · acetamide	2026-08-02 11:02	CAS(2,2)	4q	-205.274221 DMRG-SCF	-205.271013 QPU (H) (HF)	3.21 mHa · noise	21.0s QPU	i
KEAP1_LOF (mutant) · methanethiol	2026-08-01 11:36	CAS(2,2)	4q	-432.892979 DMRG-SCF	-432.888596 QPU (H) (HF)	4.38 mHa · noise	21.0s QPU	i
KEAP1_LOF (mutant) · methanethiol	2026-08-01 11:35	CAS(2,2)	4q	-432.892979 DMRG-SCF	-432.894073 QPU (H) (HF)	1.09 mHa · noise	21.0s QPU	i
TP53_C275F (mutant) · toluene	2026-08-01 10:35	CAS(2,2)	4q	-266.472783 DMRG-SCF	-266.468676 QPU (H) (HF)	4.11 mHa · noise	21.0s QPU	i
KEAP1_LOF (mutant) · methanethiol	2026-07-24 11:28	CAS(2,2)	4q	-432.892979 DMRG-SCF	-432.897300 QPU (H) (HF)	4.32 mHa · noise	21.0s QPU	i
TP53_C275F (mutant) · toluene	2026-07-24 08:40	CAS(2,2)	4q	-266.472783 DMRG-SCF	-266.469870 QPU (H) (HF)	2.91 mHa · noise	21.0s QPU	i
TP53_C275F (native) · methanethiol	2026-07-19 16:27	CAS(2,2)	4q	-432.892976 DMRG-SCF	-432.897146 QPU (H) (HF)	4.17 mHa · noise	21.0s QPU	i
TP53_C275F (native) · methanethiol	2026-07-19 16:14	CAS(2,2)	4q	-432.892976 DMRG-SCF	-432.892976 QPU (H) (HF)	0.00 mHa · noise	—	f
TP53_C275F (mutant) · toluene	2026-07-16 08:13	CAS(2,2)	4q	-266.472783 DMRG-SCF	-266.472617 QPU (H) (HF)	0.17 mHa · noise	21.0s QPU	i
TP53_C275F (native) · methanethiol	2026-07-15 16:19	CAS(2,2)	4q	-432.892976 DMRG-SCF	-432.900181 QPU (H) (HF)	7.21 mHa · noise	21.0s QPU	i
TP53_C275F (native) · methanethiol	2026-07-15 15:01	CAS(2,2)	4q	-432.911995 DMRG-SCF	-432.892976 QPU (H) (HF)	19.02 mHa · noise	—	f
H2_smoketest (native) · H2	2026-07-15 09:30	CAS(2,2)	4q	-1.857273 DMRG-SCF	-1.836944 QPU (H) (HF)	20.33 mHa · noise	21.0s QPU	i
H2_smoketest (native) · H2	2026-07-15 09:12	CAS(2,2)	4q	-1.857273 DMRG-SCF	-1.836966 QPU (H) (HF)	20.31 mHa · noise	—	f

✓ = P8 seal re-verified & notarized by **LEON** at ingestion · "Verify" re-checks the seal live

Screenshot 4. SOLANGE platform, Compliance: the LEON audit trail and the P1 to P9 provenance schema (Cohen, 2026a).

QUALITY / REGULATORY VIEW

Compliance & Audit Trail

FDA 21 CFR Part 11 · Reproducibility Engine · Provenance Metadata

✓ **297 notarization events** recorded in LEON's append-only audit trail — insert-only, no update or delete path exists.

✓ **Provenance Status: SEALED** — All 9 provenance elements (P1–P8 + P9 ML-decoder) captured · FDA 21 CFR Part 11 audit-trail requirements *aligned* (not a certified-compliant system) · Run ID: 714288a4-92f6-4a7b-8fb7-c016fbc4f603 · Phase 3A LIVE · 9/25/2026, 4:21:19 PM

AUDIT TRAIL — FULL RUN LOG

4:21:19 PM ● **Run initiated** — Target: N2 pipeline self-test · Mutation: N2 · PDB: null · Run ID: 714288a4-92f6-4a7b-8fb7-c016fbc4f603 · Phase 3A PennyLane simulator

+0s ● **Active space defined** — 4e / 4 orbitals · Encoding: Jordan-Wigner (PySCF CAS(4e,4o) → openfermion → 329 Pauli terms) · Basis: STO-3G (PySCF CASSCF(4,4))

+0s ● **Circuit compiled** — Ansatz: AllSinglesDoubles UCCSD (4e/8q, PennyLane HF initial state) · Qubits: 8 · Gates: 27 · Depth: 27 · Hash: 64f913f686f5c29d...

+175.947s ● **VQE converged** — 80 iterations · Active-space energy: -5.6832588596999996 H_a · Variance: 8.471e-9 · Backend: default.qubit

+175.947s ● **Total energy assembled** — Core (frozen) energy: -101.91463698341504 H_a + Active-space: -5.6832588596999996 H_a = -107.59789584311504 H_a · CASSCF reference: -107.59836731010016 H_a · Model compound: n2_benchmark

+175.947s ● **P7 statistical estimator applied** — Energy: -107.59789584311504 H_a · CI: [-107.59795289, -107.59783880] · Confidence: 95% · Method: Bootstrap CI over last 20 optimisation steps

+175.947s ● **P8 cryptographic seal applied** — Algorithm: SHA-256 · Seal: 726f7d0f16e8c9... · All provenance elements locked · FDA 21 CFR §11.10(e) audit trail-aligned

P1–P9 PROVENANCE SCHEMA

ELEMENT	DESCRIPTION	LAYER	STATUS
P1	Circuit fingerprint — ansatz type, gate count, depth, qubit map	QC Kernel	✓ Captured
P2	Compilation lineage — transpiler version, SWAP overhead, native gate set	Hybrid	✓ Captured
P3	Device & calibration — backend ID, calibration epoch, T1/T2/gate fidelity snapshot	QC Kernel	✓ Captured
P4	Error budget snapshot — per-gate error rates at execution time	QC Kernel	✓ Captured
P5	Raw outcome distribution — shot-level bitstring histogram before mitigation	QC Kernel	✓ Captured
P6	Error-mitigation lineage — ZNE noise factors, Pauli twirling config, mitigation version	Hybrid	✓ Captured
P7	Statistical estimator + 95% CI — mitigated energy, bootstrap interval, GNN-corrected value	Hybrid	✓ Captured
P8	Cryptographic seal — SHA-256 hash chain P1+P2→...→P7+P8 — tamper-evident seal	Governance	✓ Sealed
P9	ML-decoder lineage — Nvidia Ising 3D CNN model ID, weight checksum, training corpus hash, inference config (conditional, present when QEC uses a learned ML decoder)	Hybrid	✓ Sealed (minimal)

Screenshot 5. SOLANGE research poster, IST 697, May 2026.

CENTER FOR INFORMATION SYSTEMS & TECHNOLOGY
Claremont Graduate University

AI for Humanity Lab
Claremont Graduate University

Doron Cohen, Doctoral Candidate
Supervised by Prof. Itamar Shattat

Deciding When Quantum Computation Is Warranted: A Governed Instrument for Classical-Tractability Adjudication in Non-Druggable Oncology

A Design Science approach to translating quantum-computed molecular evidence into governed, reproducible, clinically-accessible knowledge.

solange-platform.bio

THE PROBLEM

Non-druggable TP53 mutations exceed classical computational chemistry.

~60% of NSCLC patients have no FDA-approved targeted therapy. TP53 structural mutants like p.Gln279Phe (C275F) disrupt the DNA-binding interface at the loop-sheet-helix region — no exploitable cavity for conventional drug design.

- DFT (B3LYP): 2–4 kcal/mol error → AG shifts x5–6 → candidate misclassification
- CCSD(T): gold standard, but O(N⁷) infeasible at 48 active electrons
- FCI-QC[2] → 10¹⁷ ops at N=48 — intractable

BQP-relevance: the fermionic Hamiltonian eigenvalue problem at C275F admits a known BQP-class quantum algorithm. Real DMRG measurements on today's generic, not-yet-chemis-defined active space, classify Class B (quantum-enhanced) across every target tested. A chemis-defined setup could plausibly shift a target to Class A (quantum-necessary). The architecture is built for either outcome.

THE SOLUTION — THREE-LAYER ARCHITECTURE

NGS PATIENT INPUT Patient case M25-0349 · TP53 C275F · non-druggable tumor-suppressor mutation

- LAYER 1 — BQP-RELEVANCE CLASSIFICATION** (classical)
- BQP-relevance classification framework · non-druggable tumor-suppressor mutations
- Translational orchestration · oncology-specific · Part-11-governed · bifurcation-aware
- FDA 21 CFR Part 11 framework with ML-decoder provenance, P1–P9
- LAYER 2 — HYBRID INTEGRATION SUBSTRATE** (hybrid)
- HPC orchestration · pull-based agent, USC CARC "Laguna" · no stored credentials
- Active-space selection · AVAS (PySCF), RDKit-built fragments from real PDB coords
- Classical-tractability classifier · block2 DMRG, J&J 2.6 mHa convergence
- Quantum execution · PennyLane VQE (3A) · IBM Heron r3 (3B)
- LAYER 3 — QUANTUM KERNEL** (scientific workflow)
- IBM Heron r3 · 154 qubits · heavy-hex lattice · improved gate/readout fidelity (IBM Quantum 2026, v3 km_pittsburgh)
- sqDRIFT · sample-based quantum digitalisation (theoretical, stochastic drift)
- Empirical anchor · 32-electron / 300-qubit C₆₀ benchmark (IBM Research 2026, arXiv:2603.08696)
- IBM roadmap · Starting 2029: 200 logical qubits · C275F (24–56q) within trajectory

THE PROTOTYPE

Live, deployed, navigable

Live QC-AI-HPC platform on Netlify · anchored in real NGS data (M25-0349), 3D molecular viewer · 8 DMRG-classified targets · six-screen workflow.

- 8 DMRG-classified targets · TP53 C275F, R175H, G245S, R249S, R282W, KEAP1 G333C, STK11 D154N, SETD2 R1625C; all measured Class B.
- Mutation-aware Hamiltonian per mutation class
- P1–P9 audit-trace emission (ML-decoder lineage conditional)
- Cross-disciplinary views — oncology, pharma, hospital IT
- Bifurcation-aware backend (Qiskit live today · CUDA-Q-compatible for Heron r3)

→ solange-platform.bio

LIVE DEMO — TP53 C275F QUANTUM SIMULATION

The active site — visualized, navigable, BQP-classified.

React 3D viewer:

- TP53 C275F + candidate ligand (Eprentosyl / APR-246) shown at the loop-sheet-helix DNA-contact region (illustrative)
- Encircling rings: wavefunction visualization
- Orange field: active-space region (illustrative)
- 24–56-qubit BQP target within IBM Heron r3 capacity

METHODOLOGY

Five DSR phases. Twenty months. Triangulated evaluation.

- Phase 1 · Foundation · NGS · BQP-relevance · active-space scoping ✓ complete
- Phase 2 · Architecture · Workflow spec · ReactThree.js prototype ✓ complete
- Phase 3A · Live Circuit · PennyLane-4-qubit live VQE + 20e/40q HPC run, both complete (LEON sealed, Aug 2026) · PySCF 3W Hamiltonian ✓ complete
- Phase 3B · Heron r3 · C275F: 24–56q active site, DMRG Class B · proven feasible (94q, More et al. 2026) · KEAP1/STK11: current-era, 32–74q, DMRG Class B · PDB-verified
- Phase 4 · DSR Evaluation · 4 layers · CSF are registered · m4-e4 · Jan-Apr 2027
- Phase 5 · Synthesis & Defense · May-Dec 2027

Architecture is the contribution. Simulation is the application. Patients are the motivation.

RESULTS & BENCHMARKING — QC vs DFT vs CCSD(T)

Method comparison at the C275F active-space boundary.

Illustrative platform outputs:

- DFT B3LYP: ~8.4 kcal/mol (2.8 error — outside chemical accuracy)
- sqDRIFT (Phase 3B target): ±1 kcal/mol error
- CCSD(T) reference: ~11.2 kcal/mol

Benchmarking framework is itself an original IT contribution.

THE IMPACT

To our knowledge, first FDA 21 CFR Part 11-governed quantum-chemistry orchestration for oncology.

- Proposed BQP-relevance classification for non-druggable oncology targets · reusable across mutation classes
- Unlike quantum-enhanced approaches targeting classically-tractable proteins (e.g. KRAS — Vakili et al., Nature Biotechnology 2025), quantum necessity here is established empirically, not assumed — a DMRG classifier tests every target before any quantum claim is made, now cross-checked against an independent solver (SHC, 0.198 mHa agreement)
- Novel Part-11 compliance framework for probabilistic quantum binding data (incl. P9 ML-decoder provenance)
- Bifurcation-aware orchestration · CPU and GPU (NVIDIA L40S) backends both provisioned on Laguna; GPU path proven at benchmark scale, not yet production-scale
- Selective computational escalation · formal routing of subproblems to AI / HPC / QC

56
C275F's largest active-space — within Heron r3 capacity

8
DMRG-classified targets, all Class B

100%
of measured targets converged to Class B (B/R, DMRG)

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Screenshot 6. Manuscript prepared for arXiv submission, September 2026 working draft, first page (Cohen, 2026b).

A Governed QC-AI-HPC Architecture for Deciding When Quantum Computation Is Warranted in Non-Druggable Oncology

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Abstract

Regulated quantum-oncology computation is an information-systems problem before it is a computational-chemistry problem. The first question is not how to run a quantum algorithm but whether a given target warrants one. This paper presents a governed QC-AI-HPC orchestration architecture that adjudicates, per mutation target, whether quantum computation is warranted, and that records every step of that decision in a nine-element P1-P9 provenance record aligned with the audit-trail logic of FDA 21 CFR Part 11.

The architecture has three layers. Layer 1 is the quantum-kernel layer, anchored by sample-based quantum diagonalization (SqDRIFT) on IBM Heron-class hardware. Layer 2 is the hybrid-substrate layer, covering classical preprocessing, GPU/HPC services, AI calibration, and emerging learned quantum-error-correction decoders. Layer 3 is the governed orchestration and provenance layer. Its central component is a two-method classifier: DMRG and SHCI, two classical methods with independent failure modes, run on the same active space, and the platform records the verdict of each method and any disagreement between them.

The classifier has now been exercised on real targets. Every real biological target measured to date, including TP53 C275F at active spaces of 36 and 48 electrons, is classically tractable (Class B). The same classifier returns the opposite verdict (Class A) for a declared model-compound proxy of a [2Fe-2S] redox center, so the Class B findings are measurements rather than a default. Small pipeline test jobs (4 to 8 qubits) also ran on three IBM Heron-class devices in July and August 2026; they validate the pipeline, not chemical accuracy. A design-science evaluation protocol is proposed. No energy result in this paper should be interpreted as a quantum-computed molecular binding energy.

Keywords: information systems; design science research; emerging computing; QC-AI-HPC orchestration; quantum computing; high-performance computing; artificial intelligence; precision oncology; classical-tractability adjudication; DMRG; SHCI; TP53; NSCLC; SqDRIFT; IBM Heron; NVIDIA NVQLink; FDA 21 CFR Part 11; provenance.

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Screenshot 7. SOLANGE dissertation proposal, title page (Cohen, 2026a).

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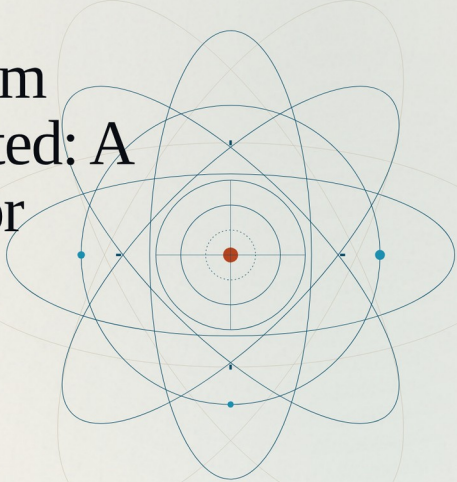
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DISSERTATION PROPOSAL

SOLANGE — SCIENTIFIC ONCOLOGY LEGACY: ADJUDICATION OF NECESSITY THROUGH GOVERNED EVIDENCE



Deciding When Quantum Computation Is Warranted: A Governed Instrument for Classical-Tractability Adjudication in Non-Druggable Oncology



A Design Science Approach to the Systems Problem of Translating Quantum Computation

Screenshot 8. SOLANGE dissertation abstract, stating that every real biological target measured to date is classically tractable (Cohen, 2026a).

Three original IT contributions are claimed: a BQP-relevant classification framework for non-druggable NSCLC targets, a Part-11-governed QC·AI·HPC orchestration architecture for oncology mutation simulation, and the first compliance framework (as of June 2026) applying FDA 21 CFR Part 11 to probabilistic quantum-computed molecular binding data — including ML-decoder provenance (P9) for emerging Ising-class learned QEC. **The contribution of this dissertation is not a computed energy but the governed instrument that adjudicates whether one is even worth computing on a quantum device** — a decision made against real DMRG measurement rather than assumed, and now cross-checked, for the first time in this line of work, against a second classical method (SHCI) whose failure mode is independent of DMRG's. Every real biological target measured to date is found classically tractable (Class B, quantum-advantaged rather than quantum-necessary) — a validated empirical finding, not an absence of results, since the same classifier is separately shown capable of returning the opposite verdict: both DMRG and SHCI independently classified a declared model-compound proxy for a [2Fe–2S] redox center (CAS(46,32)) as Class A, agreeing on the verdict even though their raw energies did not converge to the same value — itself a methodological finding about reading energy agreement and class agreement as distinct signals (§06.i). Full-scale execution on IBM Heron r3 quantum hardware is identified as a subsequent research phase, enabled by the architecture designed here.